DESIGN - ESTIMANDS - UNCERTAINTY - ROBUST FINDINGS

BurnBench, Specified.

A standalone statistical account of 5 models x 8 harnesses x 33 measured engineering tasks: what is directly observed, what depends on aggregation, what remains identifiable, and what should be treated as a bound or sensitivity result.

MODELS

HARNESSES

1,247 CELLS Model x harness x task	5,233 TRIALS 3-5 in most cells	83.89% STRICT PASS RATE 4,390 verified passes	25.04% MIXED CELLS 308 / 1,230 repeated	25 COMMON TASKS For fully comparable views

STATISTICAL STANCE

The analysis is descriptive of this benchmark matrix. Task bootstraps quantify sensitivity to the sampled tasks; they do not turn five models and eight harnesses into a random population sample.

00 ABSTRACT

What the data supports

ABSTRACT

BurnBench contains 1,247 model x harness x task cells and 5,233 hidden-oracle trials. The central result is stable: across observed stack means, the harness main effect exceeds the model main effect, while model x harness interaction remains large. Harness ranks transfer only partially across models. One quarter of repeated cells produce mixed outcomes, but the famous flake/pass correlation is largely mechanical. Cost varies by more than two orders of magnitude with little relation to pass rate. Easy-suite ranking has weak strict agreement with a five-task hard band, and judge calibration degrades at stack level on those tasks. All statements are conditional on task coverage, aggregation, and the cell-level export.

51.2%
HARNESS SHARE
54.1% common-25

12.4%
MODEL SHARE
15.5% common-25

0.29-0.42
RANK TAU-B
Specification range

126-161x
COST SPREAD
Weighting-dependent

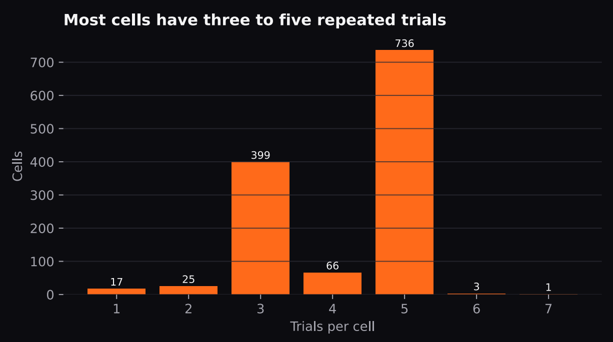
Contents

01 DESIGN AND DATA GRAIN

One row is a cell, not a trial

The shipped dataset has one row per model x harness x task. Each row contains trial counts, strict and relaxed pass counts, Wilson intervals, cell-level median token/cost/latency summaries, judge aggregates and rationales, graded-score fields, noop counts, and a non-determinism flag.

```
cell = (model, harness, task)
n_trials in {1,...,7}
strict_rate = strict_passes / n_trials
```



Trial-depth distribution: 736 cells have five trials, 399 have three, and 66 have four; 17 cells have one trial.

NOT AVAILABLE AT THIS GRAIN

Per-trial pass-linked cost, token, latency, noop overlap, timestamps, and retry sequences. Therefore actual successful-run cost and exact executed-only pass rates are not identifiable.

Verification

Hidden executable oracles run after the agent exits.

Strict passes

4,390 of 5,233 trials, or 83.8907%.

Solution passes

4,542 of 5,233 trials, or 86.7953%.

Judge coverage

4,701 judged outputs, 89.83% of runs.

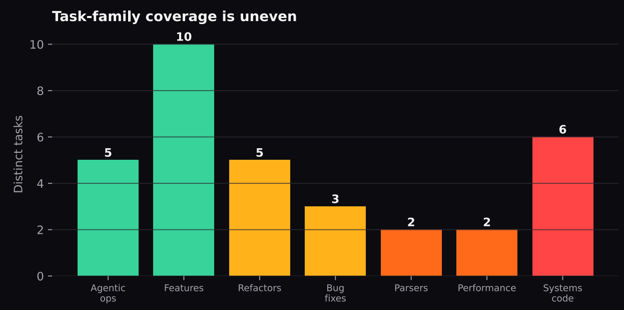
UNIT-OF-ANALYSIS RULE

Any analysis that needs trial-level joint distributions must be labeled a bound, a proxy, or unavailable. Cell aggregates cannot reconstruct which noop trial passed, which failed attempt cost more, or which latency belongs to a success.

02 COVERAGE AND TASK POPULATION

Broader scope and comparability are different goals

Task counts range from 26-28 for Qwen harnesses to 31-33 for other models.



Ten feature tasks and six systems tasks dominate the family mix; parser and performance have only two each.

33
MEASURED TASKS
Broad as-run scope

25
COMMON TASKS
All 40 stacks complete

5
HARD TASKS
Qwen absent; 3 are systems

AS-RUN ESTIMAND

Answers: how did each observed stack perform over everything it was assigned? It maximizes scope but mixes task compositions across models.

COMMON-25 ESTIMAND

Answers: how did all 40 stacks compare on the same 25 tasks? It is narrower, but it is the default for cross-model capability and value.

GENERALIZATION BOUNDARY

The benchmark describes these five models, eight harnesses, and selected tasks. Task bootstrapping addresses task-selection sensitivity within the measured universe; it does not establish a population sample of all coding work or all agent products.

03 OUTCOMES AND ESTIMANDS

Every number needs a specification

Outcomes

```
strict_rate = sum(strict_passes) /
sum(n_trials)
```

```
solution_rate = sum(solution_passes) /
sum(n_trials)
```

```
flaky_cell = I(n > 1 and 0 < strict_passes <
n)
```

```
proxy_cost_per_pass =
weighted_median(task_cost) / strict_rate
```

Required specification fields

FIELD	TYPICAL VALUES
task universe	as-run / global common-25 / hard-5 / custom
outcome	strict / solution
noop policy	as-run / best bound / worst bound
cost weighting	trial-weighted / task-equal
uncertainty	Wilson / task bootstrap / both
aggregation	cell / stack / family / pooled trials

Two layers of uncertainty

RUN UNCERTAINTY

Wilson intervals condition on a fixed cell and treat trials as independent Bernoulli draws. They describe stochastic uncertainty for that exact setup.

TASK-SELECTION UNCERTAINTY

Task-cluster bootstraps resample tasks and pool all their runs. They answer how a slice moves when the selected tasks change.

INDEPENDENCE CAVEAT

Trials share task prompt, container, endpoint, and often execution defaults. Wilson intervals can be anti-conservative under within-cell dependence. They should not be used as the sole uncertainty statement for family or leaderboard claims.

04 OBSERVED STACK MATRIX

The complete cross-product is the benchmark

Balanced within each model: strict pass rate (%)

Model-balanced strict pass rates. Each row uses tasks common to all eight harnesses for that model.

MODEL	CC	CX	DR	HE	OM	OC	PI	PA
DeepSeek V4	82.0%	90.0%	84.0%	82.7%	85.3%	72.0%	82.9%	80.1%
GLM 5.2	90.4%	84.6%	88.9%	89.7%	91.8%	58.8%	88.1%	83.5%
GPT 5.6	86.5%	89.0%	86.4%	83.1%	84.8%	83.8%	86.6%	86.2%
Muse Spark 1.2	94.2%	98.5%	96.3%	97.1%	86.8%	67.6%	87.7%	80.4%
Qwen3.7 Flash	88.5%	71.4%	89.7%	83.3%	89.7%	30.8%	94.9%	79.5%

LARGEST VISIBLE CONTRAST

Qwen moves from 29.3% with OpenCode to 94.9% with Pi as run. That is the same model under two harnesses.

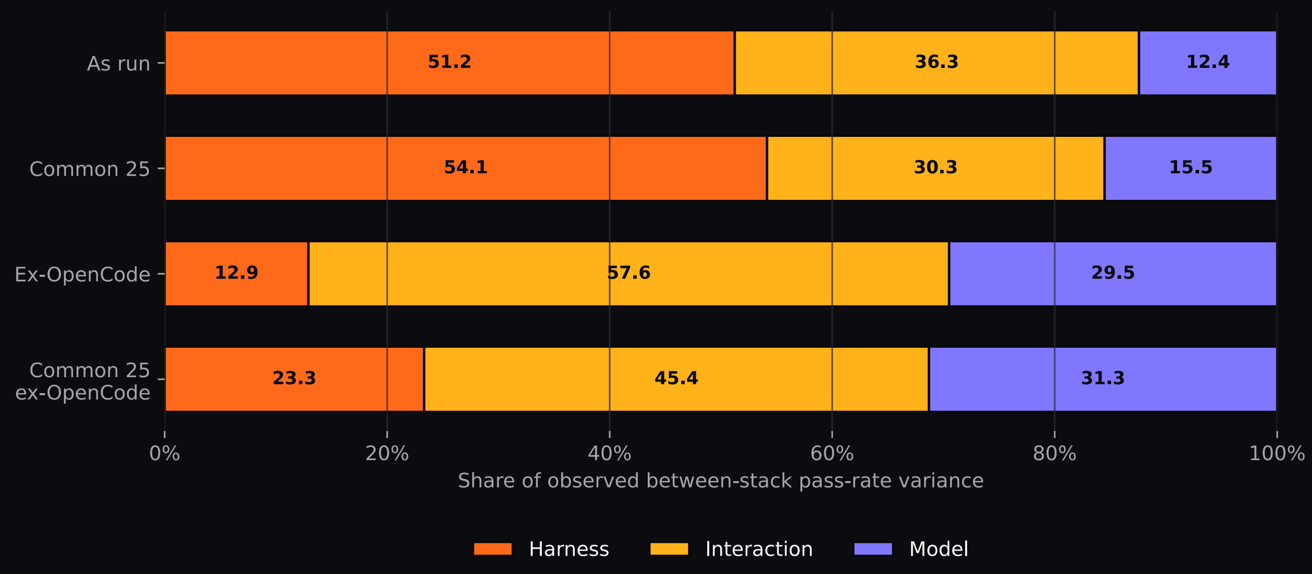
NO SINGLE GLOBAL CHAMPION

Codex leads three models, OMP leads GLM, and Pi leads Qwen. Harness identity alone is insufficient to predict rank.

05 VARIANCE DECOMPOSITION

Harness main effect exceeds model main effect

```
x_ij = stack strict pass rate
SS_model + SS_harness + SS_interaction = SS_total
```



A two-way additive decomposition of the observed 5 x 8 stack-rate matrix. No causal interpretation is implied.

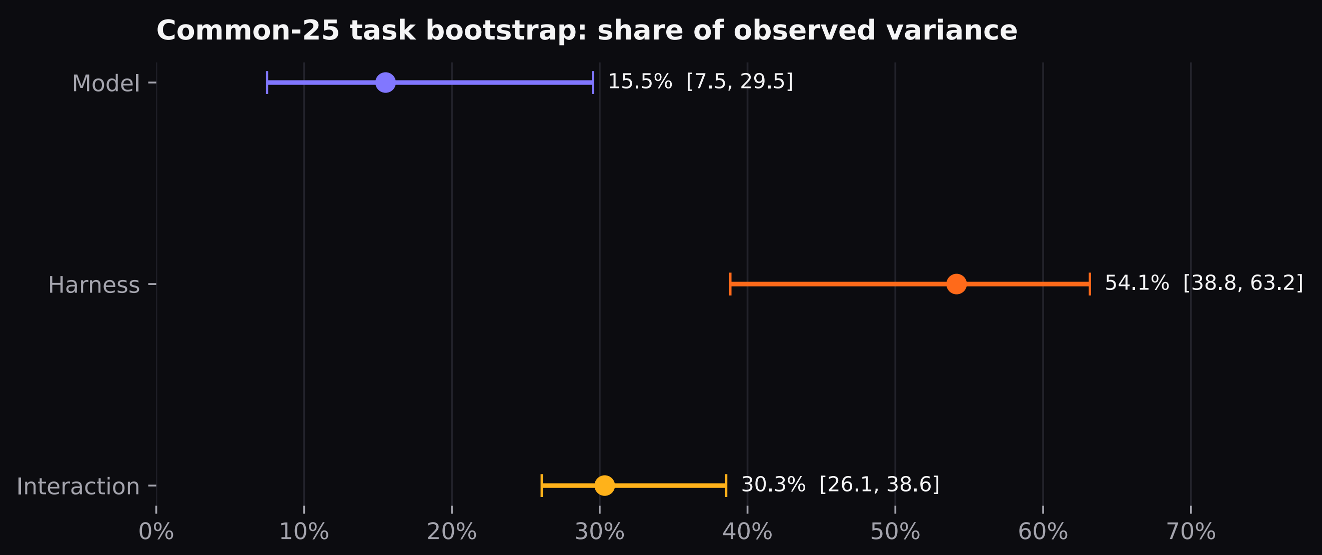
SPECIFICATION	MODEL	HARNESS	INTERACTION
As run	12.4%	51.2%	36.3%
Common 25	15.5%	54.1%	30.3%
As run, ex-OpenCode	29.5%	12.9%	57.6%
Common 25, ex-OpenCode	31.3%	23.3%	45.4%

STABLE ORDERING

With all eight harnesses, harness share is 51.2-54.1% and model share 12.4-15.5%. Excluding OpenCode, the interaction term becomes largest, at 45.4-57.6%. In every view, the stack cannot be reduced to the model.

05A VARIANCE UNCERTAINTY

Task resampling preserves the central ordering



Thirty thousand task bootstraps over the 25-task common set. Intervals reflect task selection, not model/harness population sampling.

99.5%
P(HARNESS > MODEL)
Common-25 task bootstrap

97.6%
P(INTERACTION > MODEL)
Common-25 task bootstrap

99.3%
P(HARNESS > INTERACTION)
Common-25 task bootstrap

SAMPLING-NOISE PROJECTION

A plug-in Bernoulli projection attributes about **7.3 percentage points** of total observed sum-of-squares to trial noise, including roughly 5.3 points loading into the residual interaction term.

NOISE-ADJUSTED SENSITIVITY

After subtracting the projected noise components and renormalizing: model 12.6%, harness 53.9%, interaction 33.5%. This is a sensitivity calculation, not an unbiased estimator.

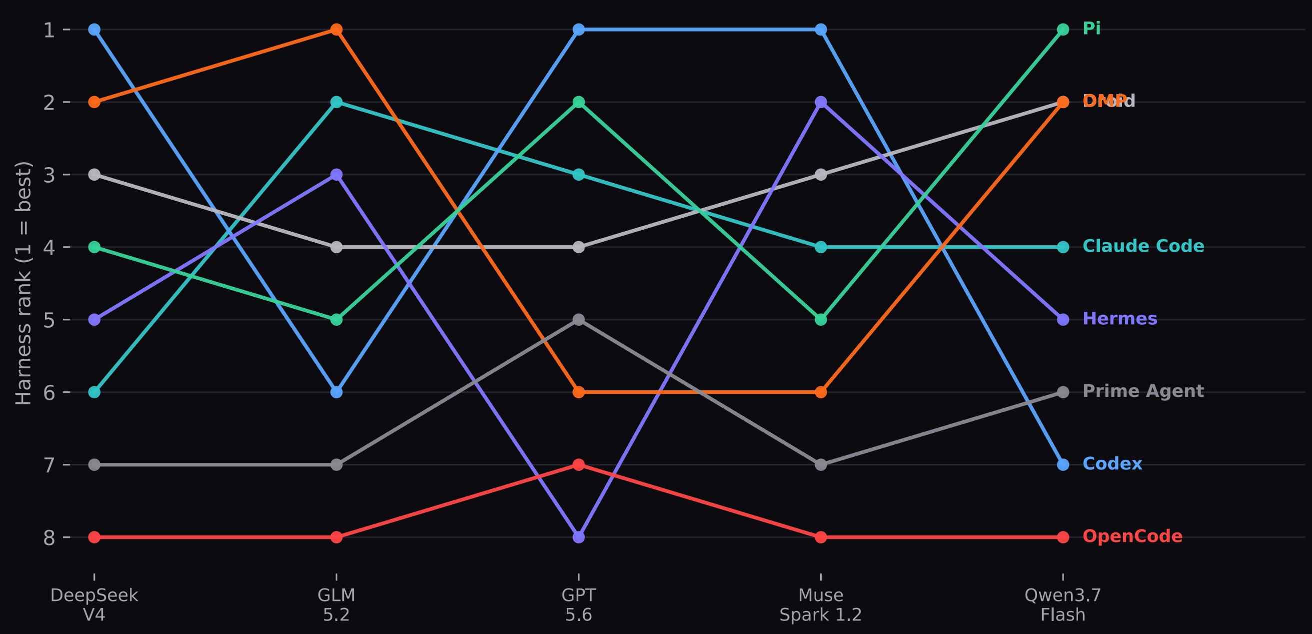
INTERPRETATION DISCIPLINE

The decomposition estimates shares of observed between-stack variance. It does not mean that switching harness causes exactly 51.2% of a future outcome, and the interaction term is an upper bound on stable pairing behavior because it absorbs residual sampling noise and configuration-specific effects.

06 RANK TRANSFER

Agreement is real, partial, and insufficient

The same eight harnesses, ranked by five models



Harness ranks are computed on each model's balanced task set. Crossing lines show model-specific ordering.

0.288
AS-RUN MEAN TAU-B
All available tasks

0.329
PAIR-BALANCED
Preferred transfer estimate

0.423
COMMON-25
Fully comparable, narrower

RESULT

Harness rankings are not random across models, but they are far from transferable. The defensible summary is **weak-to-moderate positive agreement, approximately 0.29-0.42**, not "no agreement" and not a universal harness leaderboard.

06A RANK INFERENCE AND TIES

Task balance and tie handling move the statistic

Common-25 task bootstrap

0.423
OBSERVED MEAN TAU-B
Task bootstrap 95%: 0.200 to 0.492

$p < 0.001$
PERMUTATION TEST
Independent random harness orders

The bootstrap median is lower than the original point because resampling creates additional ties and changes task leverage. The interval is descriptive of task selection.

Why one number is insufficient

CHOICE	CONSEQUENCE
As-run tasks	Broad scope, composition-sensitive; tau-b 0.288.
Pair-balanced tasks	Each model pair uses its common tasks; mean 0.329.
Global common-25	Same tasks for all 40 stacks; tau-b 0.423.
Tie-broken ranks	Produces slightly different legacy values; tau-b is preferred.

Codex
#1 for DeepSeek, GPT, Muse; #7 for Qwen.

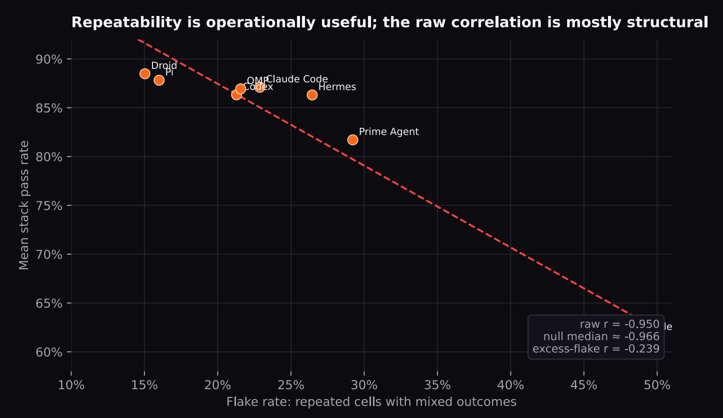
Hermes
#2 for Muse; #8 for GPT.

Pi
#1 for Qwen; #5 for Muse.

OPENCODE ASYMMETRY
Raw rank is last for four models and seventh for GPT. That consistency reflects a broad engagement failure, not evidence that every other harness has a transferable quality level.

07 REPEATABILITY PREVALENCE

One quarter of repeated cells change outcome



Raw harness-level relationship between mixed-cell rate and mean stack pass rate.

HARNESS	MIXED	FLAKE	MEAN PASS
Claude Code	35 / 153	22.9%	87.1%
Codex	33 / 155	21.3%	86.3%
Droid	23 / 153	15.0%	88.5%
Hermes	41 / 155	26.5%	86.3%
OMP	33 / 153	21.6%	86.9%
OpenCode	74 / 157	47.1%	61.7%
Pi	24 / 150	16.0%	87.8%
Prime Agent	45 / 154	29.2%	81.7%

1,230
REPEATED CELLS
n > 1

308
MIXED CELLS
Both pass and fail

25.04%
OBSERVED FLAKE RATE
Cell-level prevalence

-0.950
HARNESS-LEVEL R
n = 8 points

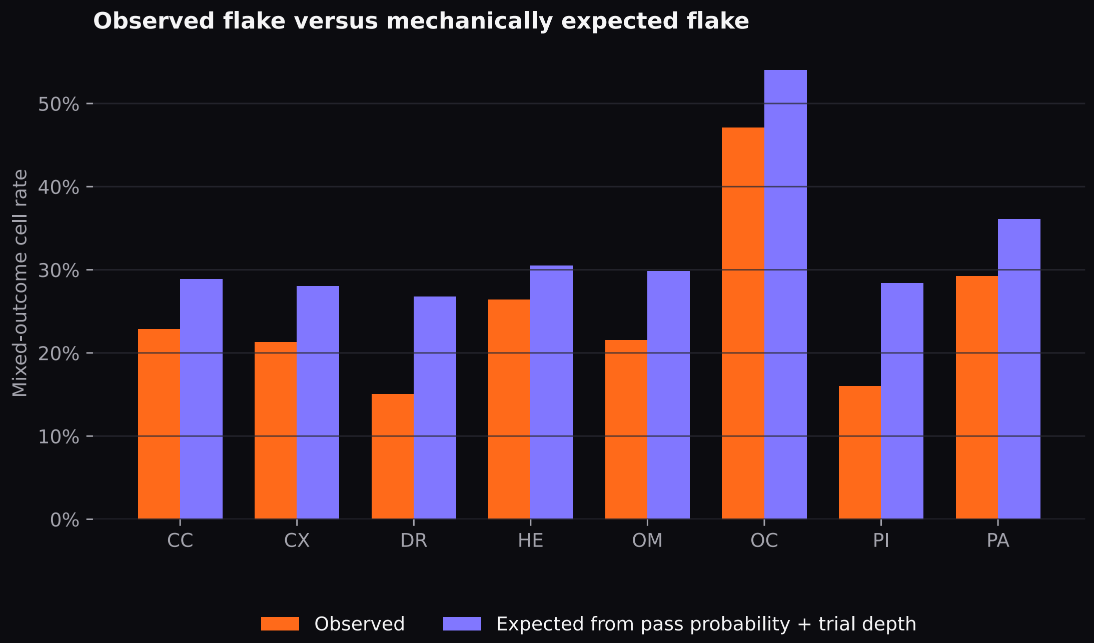
OPERATIONAL CONCLUSION

Repeated trials are indispensable. A single run does not reveal the observed 25% mixed-cell phenomenon. Report repeatability beside capability, especially for release-critical tasks.

07A MECHANICAL COUPLING

The raw -0.95 is mostly generated by the definition

```
P(flaky | p,n) = 1 - p^n - (1-p)^n
```



Expected mixed-cell rate from an additive task + model + harness binomial model with no separate reliability parameter.

-0.997	-0.966	72.5%	-0.239
EXPECTED R	NULL MEDIAN	AS/MORE NEGATIVE	EXCESS-FLAKE R
Using fitted probabilities	5,000 simulated datasets	Than observed -0.950	After expected flake removal

INVALID INFERENCE

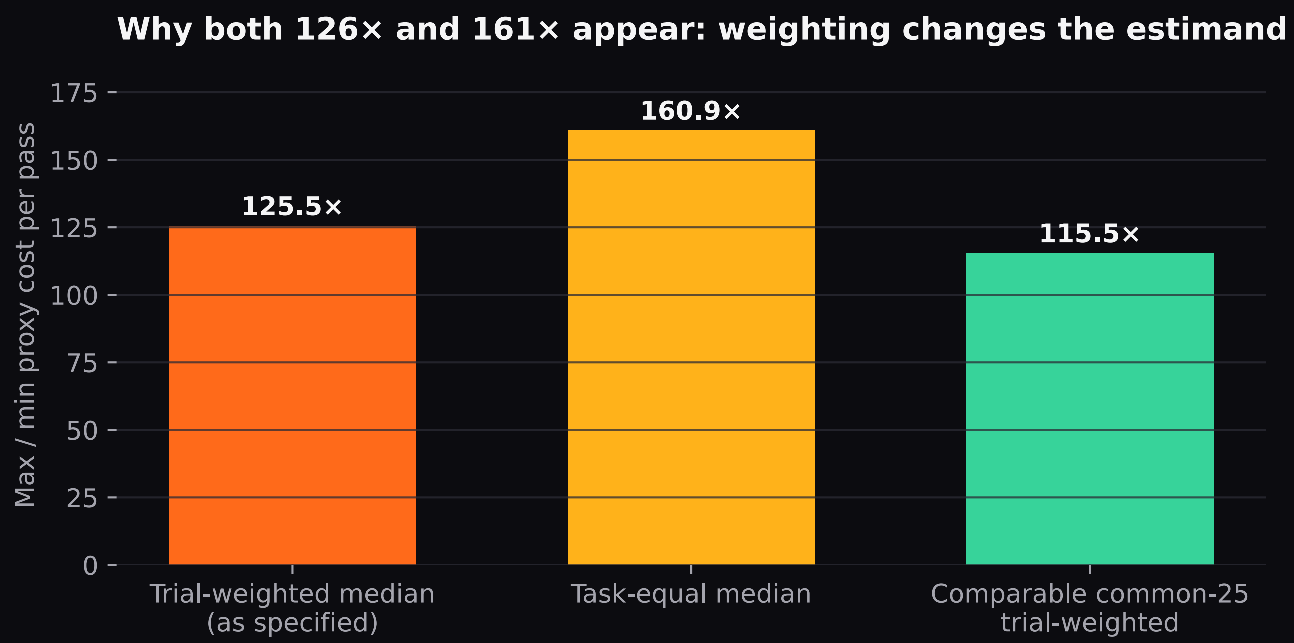
"Flakiness independently predicts capability" is not supported by the aggregate correlation. Cells at 0% or 100% pass are structurally unable to flake.

VALID INFERENCE

Repeatability remains a low-cost operational screen. Local comparisons such as Pi versus Hermes at similar mean rates suggest real harness differences that deserve targeted testing.

08 COST ESTIMANDS

126x and 161x are both correct - for different questions



The discrepancy is a weighting choice, not a contradiction in the underlying cell costs.

TRIAL-WEIGHTED MEDIAN

Tasks contribute in proportion to their trial counts. This reproduces the specified 125.52x spread: $0.002751 to $0.345303 proxy cost per pass.

TASK-EQUAL MEDIAN

Every task contributes equally regardless of trial depth. This yields 160.88x: $0.002917 to $0.469288.

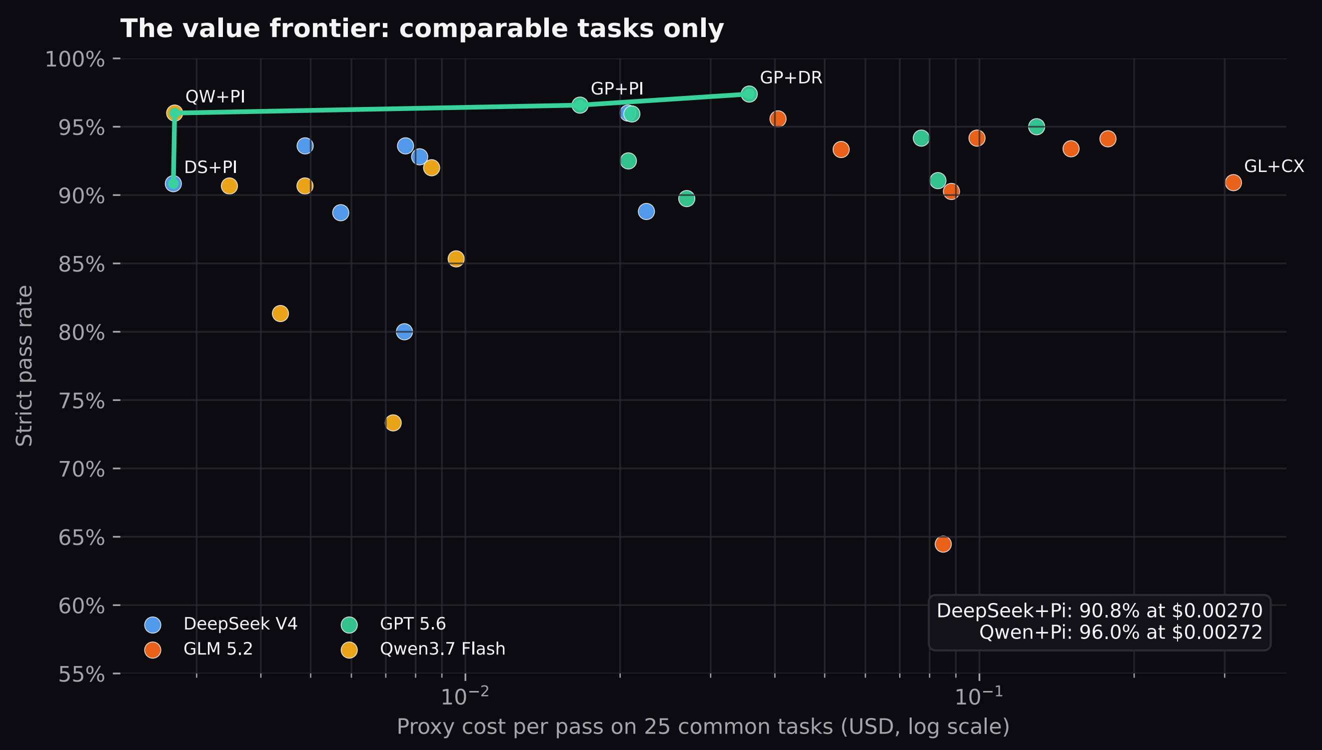
0.219 MEDIAN TASK COST X PASS p = 0.228, 32 paid stacks	0.174 PROXY COST/PASS X PASS p = 0.341	115.46x COMMON-25 SPREAD Composition-controlled

IDENTIFICATION LIMIT

"Cost per passing task" is a proxy computed from a marginal cost summary and a marginal pass rate. Without trial-level cost/outcome linkage, the actual distribution of successful-run cost and expected retry cost cannot be estimated.

08A COMPARABLE-TASK VALUE FRONTIER

DeepSeek + Pi and Qwen + Pi define the low-cost frontier



All 32 paid stacks on the 25 tasks common to every stack. X-axis is logarithmic.

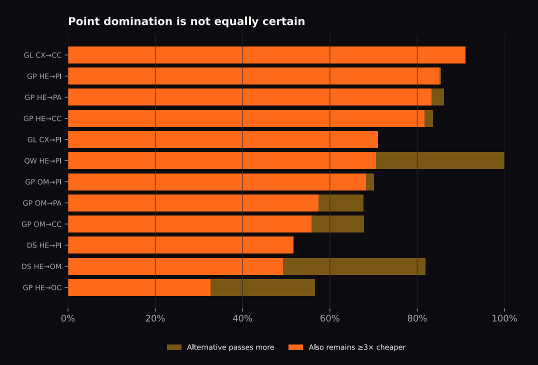
STACK, RATE >= 90%	STRICT	MEDIAN TASK COST	PROXY / PASS
DeepSeek V4 + Pi	90.83%	$0.002455	$0.002703
Qwen3.7 Flash + Pi	96.00%	$0.002610	$0.002719
Qwen3.7 Flash + Claude Code	90.67%	$0.003150	$0.003474
Qwen3.7 Flash + Droid	90.67%	$0.004419	$0.004874
DeepSeek V4 + Claude Code	93.60%	$0.004565	$0.004877
DeepSeek V4 + Droid	93.60%	$0.007158	$0.007647
DeepSeek V4 + OMP	92.80%	$0.007560	$0.008147
Qwen3.7 Flash + OMP	92.00%	$0.007908	$0.008595

FRONTIER INTERPRETATION

DeepSeek + Pi is about 0.6% cheaper per proxy pass; Qwen + Pi is 5.17 percentage points more accurate. Reporting only one as the "cheapest path above 90%" discards a material capability difference for a negligible cost difference.

08B DOMINANCE UNCERTAINTY

Twelve point relations do not have twelve equal confidence levels



Task bootstrap support for pass superiority and full >=3x domination.

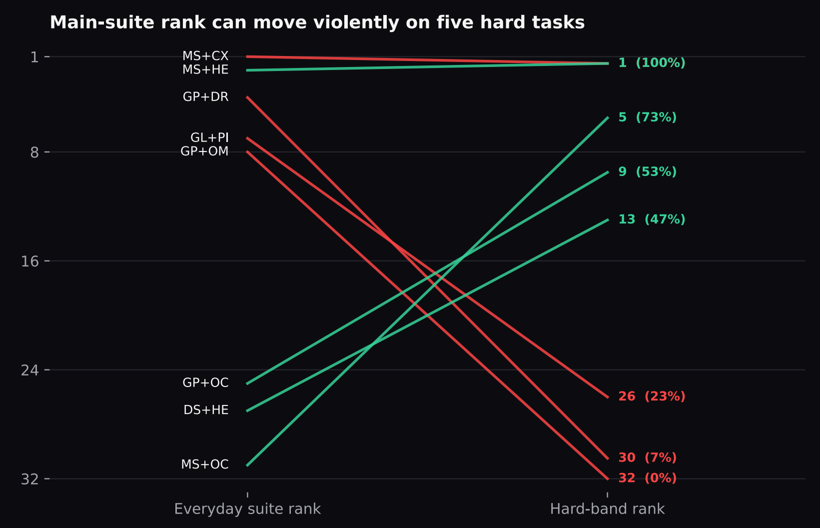
BASE → ALTERNATIVE	PAIRED GAIN	POINT RATIO	P(PASS+)	P(FULL)
GL CX → CC	+4.0 pp	5.2x	91%	91%
GP HE → PI	+3.5 pp	4.3x	85%	85%
GP HE → PA	+3.1 pp	3.9x	86%	83%
GP HE → CC	+3.4 pp	3.9x	84%	82%
GL CX → PI	+2.3 pp	7.1x	71%	71%
QW HE → PI	+11.5 pp	3.1x	100%	70%
GP OM → PI	+1.8 pp	3.9x	70%	68%
GP OM → PA	+1.4 pp	3.5x	68%	57%
GP OM → CC	+1.7 pp	3.5x	68%	56%
DS HE → PI	+0.2 pp	8.8x	52%	52%
DS HE → OM	+2.7 pp	3.2x	82%	49%
GP HE → OC	+0.7 pp	3.2x	57%	33%

INTERFACE RULE

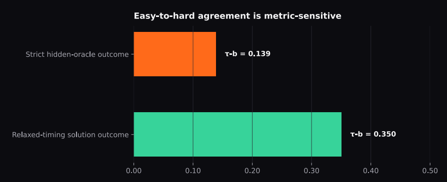
Use two labels: **point dominated** when the observed estimates satisfy the rule, and **bootstrap-supported dominated** when task resampling preserves both cheaper cost and better pass rate with a declared probability threshold.

09 SATURATION AND HARD-BAND RANKING

Strict ranking has little signal for the five hard tasks



Selected rank movements among 32 stacks. Hard-band trial depth is generally 15 per stack.



As-run main suite versus hard band.

0.086
SATURATED-BAND SD
40 stacks

0.258
HARD-BAND SD
32 stacks

STRICT OUTCOME

As-run tau-b = 0.139, p = 0.275. Common-25 task bootstrap 95%: -0.068 to 0.265.

RELAXED TIMING

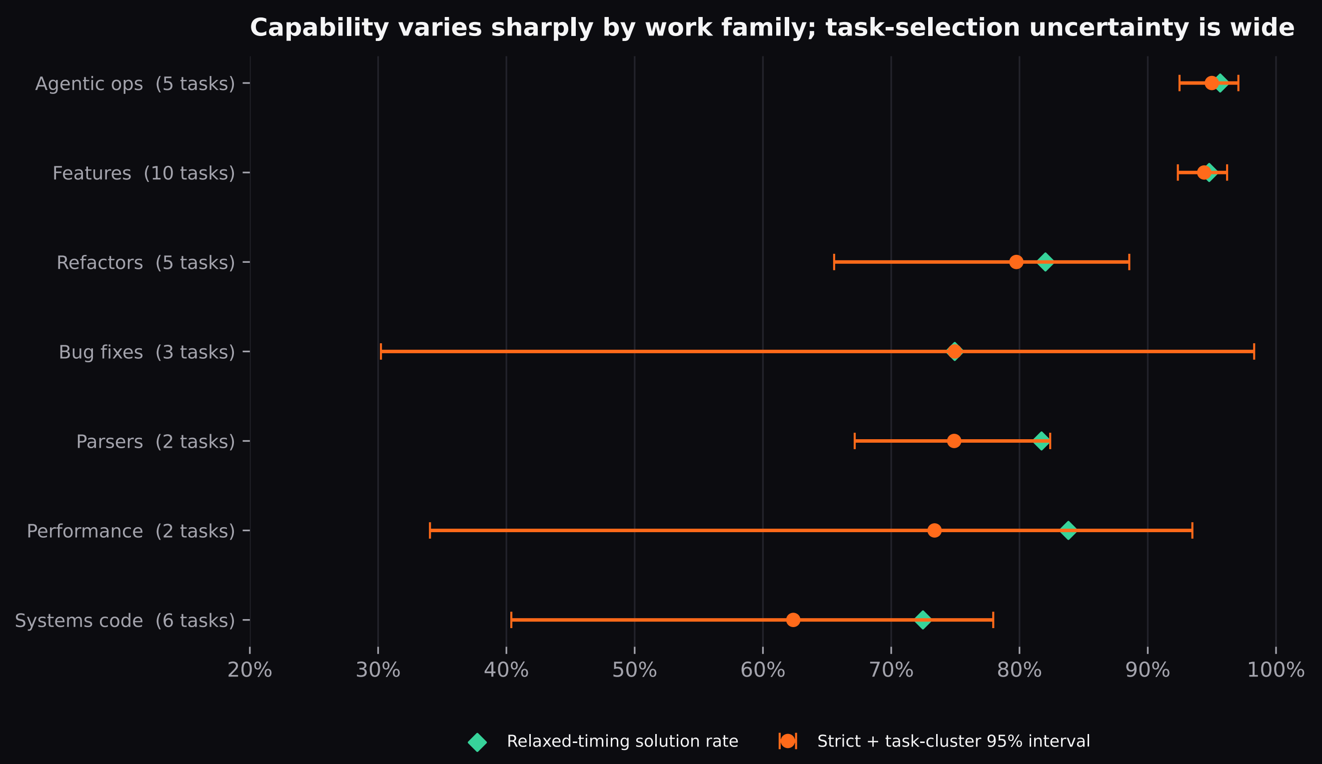
As-run tau-b = 0.350, p = 0.0069. Common-25 bootstrap 95%: -0.038 to 0.429.

LIMIT

The hard band contains five tasks, three from systems programming, and excludes Qwen. It demonstrates rank instability under this frontier slice; it is not a population estimate of all difficult software engineering.

09A CAPABILITY BY FAMILY

Run-level precision can conceal task-level uncertainty



Strict point estimates, task-cluster 95% intervals, and relaxed-timing solution rates.

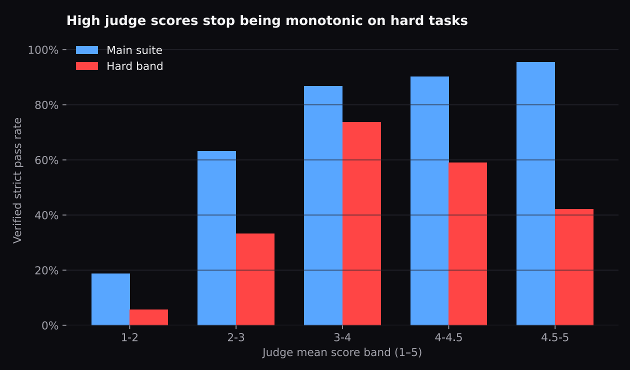
FAMILY	TASKS	TRIALS	STRICT	SOLUTION	TASK 95%
Agentic ops	5	919	95.0%	95.6%	92.5-97.1%
Features	10	1804	94.4%	94.8%	92.3-96.2%
Refactors	5	662	79.8%	82.0%	65.6-88.6%
Bug fixes	3	539	75.0%	75.0%	30.2-98.3%
Parsers	2	279	74.9%	81.7%	67.2-82.4%
Performance	2	278	73.4%	83.8%	34.0-93.5%
Systems code	6	752	62.4%	72.5%	40.4-78.0%

INTERPRETATION

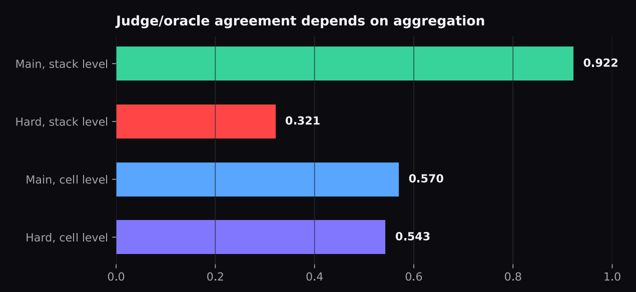
The 32.6-point gap between agentic operations and systems code is large in the measured set. The systems task-bootstrap interval is 40.4-77.8%; bugfix, parser, and performance intervals are especially wide because those families contain only two or three tasks.

10 JUDGE VALIDITY

Calibration failure is aggregation- and task-mix-specific



Verified pass rate by judge score band.



Pearson correlation between judged quality and strict outcome.

89.8%	0.922	0.321	0.570 / 0.543
JUDGED RUNS	MAIN STACK R	HARD STACK R	CELL-LEVEL R
4,701 / 5,233	n = 40	n = 32	Main / hard

COVERAGE AND DENOMINATOR ISSUE

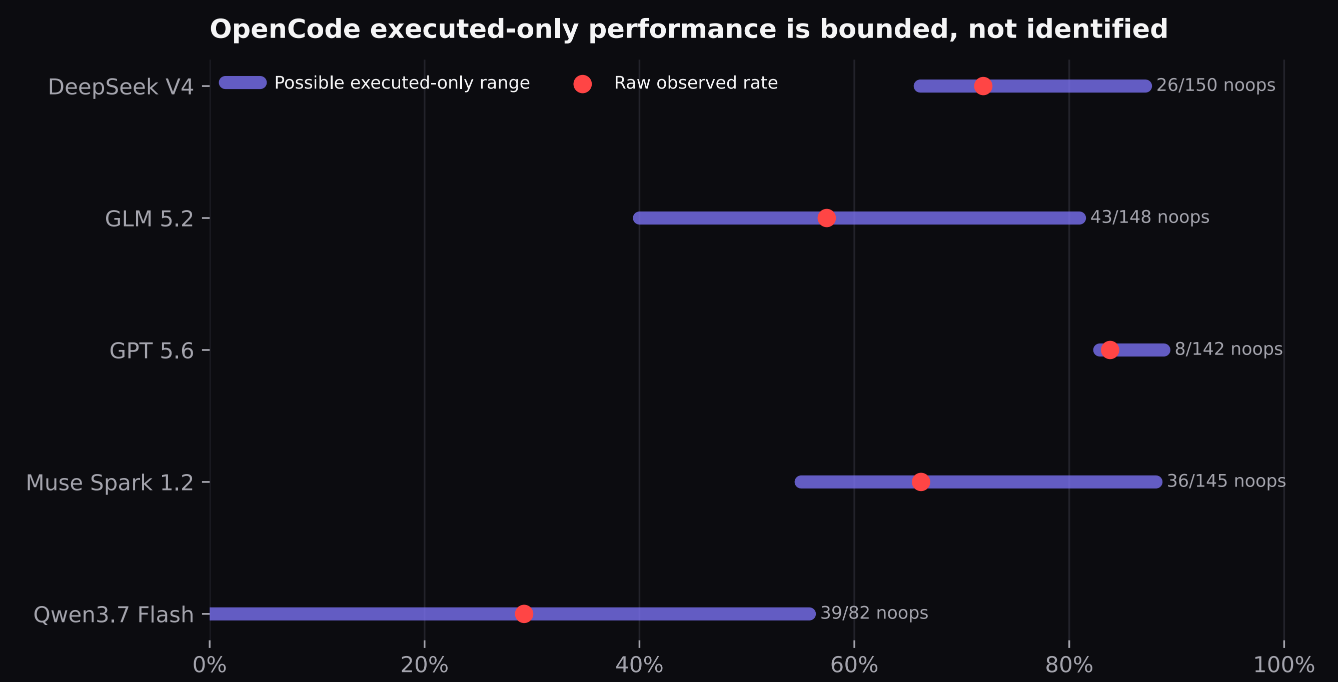
Seventy-six cells have no quality score. Ten task-03 cells have quality_n greater than retained n_trials, so pass and quality summaries are not computed over identical retained trials.

USE CASE

Judge outputs are useful for code craft and explanatory diagnostics. Executable hidden tests remain the correctness standard, especially when high-quality code can be confidently wrong.

10A OPENCODE AND NOOP BOUNDS

A start failure is still a product failure



Raw rates and mathematically possible executed-only ranges under unknown pass/noop overlap.

MODEL	PASS / TRIALS	NOOPS	RAW	EXECUTED BOUND	RAW RANK	BEST RANK
DeepSeek V4	108 / 150	26	72.0%	66.1-87.1%	8	2
GLM 5.2	85 / 148	43	57.4%	40.0-81.0%	8	8
GPT 5.6	119 / 142	8	83.8%	82.8-88.8%	7	2
Muse Spark 1.2	96 / 145	36	66.2%	55.0-88.1%	8	5
Qwen3.7 Flash	24 / 82	39	29.3%	0.0-55.8%	8	8

OBSERVED OPERATIONAL FACT

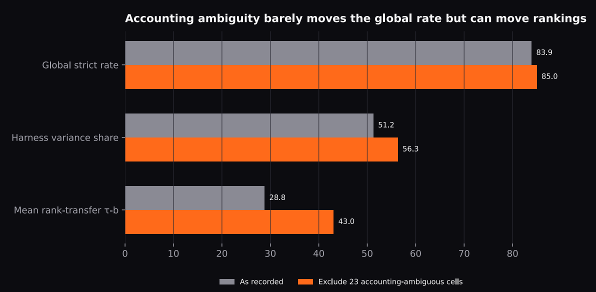
OpenCode records 152 no-tool-call runs. Qwen has 39 / 82 and GLM 43 / 148. Raw rank is last for four models and seventh for GPT.

IDENTIFICATION FACT

The cell export does not identify which passes overlap noops. Published executed-only rates use a best-case cap. The correct statistical object is a worst-to-best bound, while raw product performance remains primary.

11 INTEGRITY, LIMITS, AND CONCLUSION

The global story is stable; some rankings are not



The rank-transfer statistic is scaled by 100 in this display so all three metrics share an axis.

ACCOUNTING AMBIGUITY

23 cells covering 73 runs have zero or missing token accounting; 5 strict passes sit inside those cells. Excluding all 23 raises global strict rate from 83.89% to 84.98%.

OTHER NON-IDENTIFIABLE OR EXTERNAL CLAIMS

Trial-level joint cost/outcome data, exact noop/pass overlap, raw run logs, removed oracle cells, and the separate zero-pass IR-optimizer task are outside this CSV.

Reproducibility ledger

QUANTITY	RECOMPUTED VALUE
Cells / runs	1,247 / 5,233
Global strict / solution	83.8907% / 86.7953%
Variance, as run	M 12.424 / H 51.242 / I 36.334
Variance, common 25	M 15.515 / H 54.145 / I 30.339
Mean rank tau-b	0.288 / 0.329 / 0.423
Mixed repeated cells	308 / 1,230 = 25.0407%
Flake x pass r	-0.950077
Cost spread	125.520x / 160.878x
Strict main-hard tau-b	0.138606
Solution main-hard tau-b	0.350311
Judge stack r	0.921537 / 0.320740
Judge cell r	0.569588 / 0.542615

SURVIVING CORE

Benchmark the stack, not the model. Rank transfer is partial, repetition is essential, cost must be specified, saturated tasks should not anchor leaderboards, and executable verification remains necessary for hard work. These conclusions survive the defensible re-specifications documented in this report.

Primary file: burnbench-consolidated.csv - SHA-256 289d9695757c189d0e2a8038c0755c70b1c4bd208c4aa1c5fb6d548f9b570849 - generated from the August 2026 artifact set.